QUESTION STARTER CHEAT SHEET

	WHEN THE QUESTION SAYS...	WHAT YOU NEED TO DO
1	Complete the model by...	**ADD** the missing information to the model.
2	Identify the change in relative...	**STATE** what increased, decreased, became greater, or became less.
3	Use the model to compare the temporal scale...	**COMPARE THE TIME** each process takes. Which is faster, slower, shorter, or longer?
4	Identify the process...	**NAME** the process that is happening.
5	How does ... affect ...?	**DESCRIBE THE EFFECT.** Tell what happens to one thing when the other changes.
6	Using (...), construct an explanation for why...	**EXPLAIN WHY** it happens using information from the source given.
7	Construct an argument based on evidence that describes... (especially about cause/effect)	**MAKE A CLAIM + SUPPORT IT WITH EVIDENCE.** Show the cause-and-effect connection.
8	Justify ... by providing evidence from ...	**SHOW WHY YOUR ANSWER IS CORRECT** using evidence from the source.
9	Use evidence from ... to describe...	**DESCRIBE WHAT HAPPENS + INCLUDE EVIDENCE** from the source.
10	Which explanation about ... provides evidence for ...?	**CHOOSE** the explanation that has evidence supporting the idea.
11	Describe how ... would affect...	**TELL WHAT WOULD CHANGE** and how.
12	Which statement/ explanation/ model/ claim best/most accurately/ correctly describes/ identifies/ represents/ summarizes...	**CHOOSE THE MOST ACCURATE ANSWER** based on the information given.
13	Make an evidence-based forecast...	**PREDICT WHAT WILL HAPPEN + USE EVIDENCE** to support your prediction.
14	Which ... would be the most accurate prediction for...	**CHOOSE WHAT IS MOST LIKELY TO HAPPEN NEXT** based on the information.
15	Identify the numerical values...	**FIND AND STATE THE NUMBERS** being asked for. Include units if needed.
16	Which ... supports the claim by...	**CHOOSE THE EVIDENCE** that best supports the claim.
17	Explain how the ... supports the claim.	**CONNECT THE EVIDENCE TO THE CLAIM.** Tell why that evidence proves or supports it.
18	Use evidence from ... to make a claim about...	**MAKE A CLAIM + GIVE EVIDENCE** that supports it.
19	Analyze the ... to make a claim about...	**LOOK FOR PATTERNS/RELATIONSHIPS → MAKE A CLAIM** based on what you notice.
20	Which statement, based on evidence from ..., would refute/support the claim?	**CHOOSE EVIDENCE THAT PROVES OR DISPROVES** the claim.
21	Describe the economic benefit...	**EXPLAIN THE MONEY BENEFIT.** How does it save money, make money, create jobs, or reduce costs?
22	Discuss how the wants and needs of society are affected by...	**EXPLAIN HOW PEOPLE ARE AFFECTED.** Think about what people NEED and what they WANT.

NAME: _____ PERIOD:_____

The LANGUAGE of SCIENCE

QUICK KEY

NAME IT	DESCRIBE IT	COMPARE IT	EXPLAIN IT	JUSTIFY IT
Give the answer	Tell what happens	Tell how they are alike / different	Tell why / how	Answer + show why you are right
EVIDENCE	**DESCRIBE IT**	**PREDICT / FORECAST**	**ANALYZE**	**USE EVIDENCE THINK CRITICALLY ANSWER COMPLETELY!**
Use specific information from the source	State what you think the evidence shows	Tell what will probably happen next	Look for patterns, relationships, or important details	

SECTION 1: SCIENCE & ENGINEERING PRACTICES

PHENOMENA

What is a Phenomenon?

A **PHENOMENON** is something that happens in the _____ world that we observe, investigate, and explain.

Topic vs Phenomenon

TOPIC	PHENOMENON
A general idea or subject. Example: _____	A specific event or pattern we observe & ask questions about. Example: _____ _____ _____

WHAT SCIENTISTS DO

SCIENCE & ENGINEERING PRACTICES (SEPS)

Science & Engineering Practices (SEPs) are the actions and skills scientists and engineers use to investigate problems and explain _____

SEPS = THE

THE 8 SCIENCE & ENGINEERING PRACTICES

- ☆ Asking questions & defining problems
- ☆ Developing and using models
- ☆ Analyzing & interpreting data
- ☆ Constructing explanations
- ☆ Using mathematics & computational thinking
- ☆ Engaging in argument from evidence
- ☆ Obtaining, evaluating, & communicating information
- ☆ Designing solutions

ANALYZE VS EVALUATE

Examine information to find:	Judge how good, useful, accurate or _____ something is
_____ _____ _____ _____	

CLAIM EVIDENCE REASONING

CLAIM = Your _____ or conclusion.

EVIDENCE = Data, _____ or facts that support your claim.

REASONING = Explains HOW the evidence supports your _____

REMEMBER
A strong, scientific answer uses:
CLAIM + EVIDENCE + REASONING

CONSTRUCT VS JUSTIFY

CONSTRUCT= _____ explanation or argument using evidence & reasoning

BUILD IT FIRST → THEN JUSTIFY IT

JUSTIFY= _____ your answer is correct using evidence.

DEVELOP A MODEL

Create or improve a _____ that shows how something works

MODEL = SHOW IT

PREDICT / FORECAST

Use patterns, data, or a model to determine what is _____

PREDICT = WHAT HAPPENS NEXT

VARIABLES

Scientists change one thing to see how it affects another

INDEPENDENT VARIABLE
What is _____ in the investigation.

→

DEPENDENT VARIABLE
What is _____ or _____

REMEMBER: Measure the DEPENDENT variable

THAT'S HOW WE LEARN CAUSE AND EFFECT

SPACE SYSTEMS

- HS-ESS1-1 Develop a model based on evidence to illustrate the life span of the Sun and the role of nuclear fusion in the Sun's core to release energy that eventually reaches Earth in the form of radiation.
- HS-ESS1-2 Construct an explanation of the Big Bang theory based on astronomical evidence of light spectra, motion of distant galaxies, and composition of matter in the universe.
- HS-ESS1-3 Communicate scientific ideas about the way stars, over their life cycle, produce elements.
- HS-ESS1-4 Use mathematical or computational representations to predict the motion of orbiting objects in the solar system.
- HS-ESS1-7 Construct an explanation using evidence to support the claim that the phases of the moon, eclipses, tides, and seasons change cyclically.

HISTORY OF THE EARTH

- HS-ESS1-5 Evaluate evidence of the past and current movements of continental and oceanic crust and the theory of plate tectonics to explain the ages of crustal rocks.
- HS-ESS1-6 Apply scientific reasoning and evidence from ancient Earth materials, meteorites, and other planetary surfaces to construct an account of Earth's formation and early history.
- HS-ESS2-1 Develop a model to illustrate how Earth's internal and surface processes operate at different spatial and temporal scales to form continental and ocean-floor features.

EARTH'S SYSTEMS

- HS-ESS2-2 Analyze geoscience data to make the claim that one change to Earth's surface can create feedbacks that cause changes to Earth's systems.
- HS-ESS2-3 Develop a model based on evidence of Earth's interior to describe the cycling of matter by thermal convection.
- HS-ESS2-5 Plan and conduct an investigation of the properties of water and its effects on Earth materials and surface processes.
- HS-ESS2-6 Develop a quantitative model to describe the cycling of carbon among the hydrosphere, atmosphere, geosphere, and biosphere.
- HS-ESS2-7 Construct an argument based on evidence about the coevolution of Earth's systems and life on Earth.

WEATHER AND CLIMATE

- HS-ESS2-4 Use a model to describe how variations in the flow of energy into and out of Earth's systems result in changes in climate.
- HS-ES3-5 Analyze geoscience data and the results from global climate models to make an evidence-based forecast of the current rate of global or regional climate change and associated future impacts to Earth systems.
- HS-ESS2-8 Evaluate data and communicate information to explain how the movement and interactions of air masses result in changes in weather conditions.

HUMAN SUSTAINABILITY

- HS-ESS3-1 Construct an explanation based on evidence for how the availability of natural resources, occurrence of natural hazards, and changes in climate have influenced human activity.
- HS-ESS3-2 Evaluate competing design solutions for developing, managing, and utilizing energy and mineral resources based on cost-benefit ratios.
- HS-ESS3-3 Create a computational simulation to illustrate the relationships among management of natural resources, the sustainability of human populations, and biodiversity.
- HS-ESS3-4 Evaluate or refine a technological solution that reduces impacts of human activities on natural systems.
- HS-ESS3-6 Use a computational representation to illustrate the relationships among Earth systems and how those relationships are being modified due to human activity.

SECTION 5: ENGINEERING & DESIGN

ENGINEERING STARTS WITH A PROBLEM

Engineers design _____ to meet people's _____ and _____.

1. IDENTIFY THE PROBLEM
2. DETERMINE CRITERIA AND CONSTRAINTS
3. DEVELOP POSSIBLE SOLUTIONS
4. CONSIDER COSTS, BENEFITS, AND TRADEOFFS
5. DECIDE THE BEST SOLUTION

PROBLEM VS SOLUTION

PROBLEM	SOLUTION
Something that needs to be _____ or solved.	A device, system, process, or _____ designed to address a problem.
Example:	Example:
_____	_____
_____	_____

CRITERIA

WHAT DOES THE SOLUTION _____ TO DO TO BE SUCCESSFUL?

For our lake effect snow problem, the solution should:

- Keep main roads open
- Protect drivers from crashes
- Be safe for plow crews
- Work every winter

Think: What does SUCCESS look like?

Our Criteria: The solution must keep the _____ open during a 30 inch snowfall.

CONSTRAINT

WHAT _____ OUR SOLUTION?

For our lake effect snow problem, the constraints are:

- Limited budget
- A limited amount of time
- Limited available space
- Available materials and technology
- Environmental concerns

Think: What are the LIMITS?

Our Constraint: The solution cannot _____ $4 million.

CRITERIA OR CONSTRAINT?

Ask yourself: does it describe SUCCESS or a LIMIT?

Criteria= _____

Constraint= _____

DRAWBACKS & TRADEOFFS

What you _____ when choosing one option over another.

There isn't always one _____ solution.

_____: What do we gain and what do we give up when we choose one solution over another?

COST VS BENEFIT

Engineers compare the _____ and _____ of possible solutions.

ASK: ARE THE BENEFITS _____ THE COSTS?

A larger plow fleet costs more every winter, but if it keeps ambulances moving during a storm, is the additional cost worth the benefit?

WANTS AND NEEDS OF SOCIETY

Engineering decisions are also influenced by what people _____ & _____.

NEEDS	WANTS
Things _____ for safety and well-being.	Things people _____ or would like to maintain.

THE CHALLENGE: A solution may meet one need or want while negatively affecting another.

HOW DO ENGINEERS DECIDE WHICH SOLUTION IS BEST?

They consider:

[ ] + [ ] + [ ]

+ [ ] + [ ]

Engineers weigh all of these factors to choose the solution that best meets the _____.

SECTION 2: CROSSCUTTING CONCEPTS + SCALE

WHAT ARE CCCS?

Big ideas that help us make sense of many different _____.

SEP OR CCC?

SEP = What I _____

CCC= How I _____

CCCS = THE _____

What should you look for?

- What keeps _____?
- What caused this _____?
- How _____? How _____?
- What parts are _____?
- What is moving or being _____?
- What stays the _____?
- What _____?

PUTTING IT ALL TOGETHER

PHENOMENON
What are we trying to figure out?

SEP
What will we DO to investigate it?

CCC
What kind of thinking should we use?

EXPLANATION
What does the evidence tell us?

SCALE

Scientists study Earth at very different _____ and _____ periods.

How _____? How _____?
Where?

It describes _____ or _____.

We are measuring the SIZE of the canyon.

How _____? How _____?
In what order?

It describes _____ and _____.

We are measuring the time it took to form.

SPATIAL SCALE

The size or area at which something is observed, such as how big, how far, or where it is.

- **GLOBAL** → The whole _____.
- **CONTINENTAL** → Large areas, like _____
- **REGIONAL** → Smaller regions like states or _____.
- **LOCAL** → Small areas like a _____.
- **MICROSCOPIC** → Tiny things you can only see with a _____.

TEMPORAL SCALE

The amount of time over which something happens, including how long it takes or how fast it occurs.

- **SECONDS** → Very _____ events.
- **YEARS** → Changes that happen over _____.
- **THOUSANDS OF YEARS** → Slow changes on Earth's _____.
- **MILLIONS OF YEARS** → Big changes like _____.
- **BILLIONS OF YEARS** → Very slow changes like _____.

SECTION 3: SYSTEMS & SYSTEM MODELS

WHAT IS A SYSTEM?

A group of _____
that forms a _____

☆ **KEY IDEA:**

A change in _____ part of a system can affect _____ parts.

COMPONENTS & BOUNDARIES

COMPONENTS = An _____ part of a system

BOUNDARIES = The _____ that separates the system from its surroundings.

QUESTIONS TO ASK:

- What are the_____ ?
- What is _____the system?
- what is _____the system?
- How are the parts_____?

FLOWS & FLUXES

FLOWS

The _____ of matter or energy from one place to another.

EXAMPLE MATTER FLOW:

_____

_____

EXAMPLE ENERGY FLOW:

_____

_____

- What is moving?
- Where is it going?

FLUXES

The _____ of that movement, meaning how much moves over a certain amount of time or area.

EXAMPLE MATTER FLUX:

_____

_____

EXAMPLE ENERGY FLUX:

_____

_____

- How much is moving?
- How fast?

FEEDBACK SYSTEMS

A _____ causes _____ that eventually influence the _____ change.

They describe what happens to the change.

Positive ≠ good Negative ≠ bad

POSITIVE FEEDBACK

_____ change

NEGATIVE FEEDBACK

_____change / helps stabilize

POSITIVE FEEDBACK EXAMPLE

① Temperature _____
② Ice _____
③ Less sunlight is _____
④ More energy is_____
Temperature_____ even more ④
⑤

NEGATIVE FEEDBACK EXAMPLE

① The room gets_____
② The thermostat _____ the change
③ Cooling turns ____
The temperature _____

SECTION 4: STABILITY AND CHANGE

Earth's systems are _____ .

They are constantly_____.

- What stays _____?
- What _____?
- What caused the _____?
- How does the system _____?
- Can the system _____ to its previous _____?

Earth can be constantly changing while some parts remain relatively _____.

EQUILIBRIUM

A state of _____ in a system with no overall or net _____ .

A system can be in equilibrium even when things are still _____ inside it.

DYNAMIC

Means constantly_____.

A system can be **DYNAMIC** and in **EQUILIBRIUM**

REVERSIBLE

A change where the system _____ return to its original state.

CAN GO BACK

Example of reversible:

_____

IRREVERSIBLE

A change where the system _____return to its original state.

CANNOT GO BACK

Example of irreversible:

_____

_____ ASK YOURSELF: CAN THE SYSTEM RETURN TO THE WAY IT WAS BEFORE ?

PERTURBATION

A temporary _____ that moves a system away from its normal _____ or path.

NORMAL STATE → PERTURBATION → SYSTEM RESPONDS

EXAMPLES:
- volcanic eruption
- major storm
- wildfire
- flood

Think: Something disturbed the system

THRESHOLD

The critical _____ in a system where a small additional change can cause a large, abrupt, or irreversible _____ in its state.

MORE RAINFALL → SOIL GETS WETTER → MORE RAINFALL → THRESHOLD REACHED → LANDSLIDE

A small additional change can sometimes produce a BIG response.

ANOMALY

An observation or_____ that is very different from what is expected.

- May not fit the normal_____
- May not match a _____
- May not fit a scientific _____
- May not fit baseline _____

Think: That doesn't fit the pattern!

_____ AN ANOMALY DOES NOT AUTOMATICALLY MEAN THE DATA IS WRONG.